

PHASE 3 — DOCKER & CONTAINERIZATION

DAY 11 — DOCKERFILES

Tool: Docker / Dockerfile

1. Objectives

The objectives of Day 11 are:

- Understand **base images**
- Understand the **COPY** instruction
- Understand the **RUN** instruction
- Understand **CMD / ENTRYPOINT**
- Understand and implement **non-root users**
- Understand and use **.dockerignore**

2. Theory

2.1 Base Images

A base image provides the starting environment for building a Docker image.

For this exercise, the following base image was used:

```
FROM python:3.12-slim
```

The python:3.12-slim image provides a lightweight Python environment for running the web application.

2.2 COPY

The COPY instruction copies files from the Docker build context into the Docker image.

```
COPY app.py .
```

This copies app.py into the application's working directory inside the image.

2.3 RUN

The RUN instruction executes commands while the Docker image is being built.

In this project, RUN was used to create the application user and configure file ownership.

```
RUN useradd --create-home appuser
```

```
RUN chown -R appuser:appuser /app
```

2.4 CMD / ENTRYPOINT

CMD specifies the default command that runs when a container is started.

This project uses:

```
CMD ["python", "app.py"]
```

This starts the Python web application when the container runs.

ENTRYPOINT can also be used to define the main executable for a container. In this project, CMD was used.

2.5 Non-Root Users

Running applications as a non-root user reduces the privileges available to the application inside the container.

The Dockerfile creates an application user:

```
RUN useradd --create-home appuser
```

and switches to that user:

```
USER appuser
```

Therefore, the application is configured to run as a non-root user.

2.6 .dockerignore

The .dockerignore file specifies files and directories that should not be included in the Docker build context.

The project uses:

```
__pycache__
```

```
*.pyc
```

```
.git
```

```
.gitignore
```

```
README.md
```

This helps reduce unnecessary Docker build context and prevents unrelated files from being sent to the Docker build process.

3. Hands-on Implementation

3.1 Create the Small Web Application

A simple Python web application was created in app.py.

The application listens on port 8000 and displays:

Dockerfile Web App

A terminal window screenshot from a VirtualBox environment. The prompt is 'shayan@shayan-VirtualBox:~/day11-dockerfile\$'. The command 'ls -la' has been executed, showing the following output:

```
total 16
drwxrwxr-x  2 shayan shayan 4096 Aug 31 16:11 .
drwxr-x--- 22 shayan shayan 4096 Aug 31 15:53 ..
-rw-rw-r--  1 shayan shayan  435 Aug 31 15:54 app.py
-rw-rw-r--  1 shayan shayan  174 Aug 31 16:11 Dockerfile
```

The cursor is at the end of the prompt line.

3.2

Create the Dockerfile

The Dockerfile used for the application was:

FROM python:3.12-slim

RUN useradd --create-home appuser

WORKDIR /app

COPY app.py .

RUN chown -R appuser:appuser /app

USER appuser

EXPOSE 8000

CMD ["python", "app.py"]

The Dockerfile demonstrates the required Day 11 concepts:

- Base image
- RUN
- COPY
- Non-root user
- CMD

The project directory also contains the .dockerignore file.

```
shayan@shayan-VirtualBox:~/day11-dockerfile$ ls -la
total 20
drwxrwxr-x  2 shayan shayan 4096 Aug 31 16:18 .
drwxr-x--- 22 shayan shayan 4096 Aug 31 15:53 ..
-rw-rw-r--  1 shayan shayan  435 Aug 31 15:54 app.py
-rw-rw-r--  1 shayan shayan  174 Aug 31 16:11 Dockerfile
-rw-rw-r--  1 shayan shayan   45 Aug 31 16:18 .dockerignore
```

4. Build the Docker Image

The Docker image was built using:

```
sudo docker build -t day11-web:1.0 .
```

The build completed successfully and created the image:

day11-web:1.0

The Docker build also loaded the .dockerignore file and copied the required application file into the image.

```

shayan@shayan-VirtualBox:~/day11-dockerfile$ sudo docker build -t day11-web:1.0
.
[sudo] password for shayan:
[+] Building 16.3s (10/10) FINISHED                                docker:default
=> [internal] load build definition from Dockerfile                0.1s
=> => transferring dockerfile: 213B                                0.0s
=> [internal] load metadata for docker.io/library/python:3.12-slim 3.1s
=> [internal] load .dockerignore                                   0.1s
=> => transferring context: 85B                                     0.0s
=> [1/5] FROM docker.io/library/python:3.12-slim@sha256:09f7da3bc104798d 8.6s
=> => resolve docker.io/library/python:3.12-slim@sha256:09f7da3bc104798d 0.1s
=> => sha256:48579d87405c21ba2599e244fcf3a02e900dbf9b9103e4a 250B / 250B 0.3s
=> => sha256:b79f58b3eb945fe861ace4fc38504483c5b100c79 12.12MB / 12.12MB 5.9s
=> => sha256:a0d8783128af79059fe015e6313ce9e8fd7856b194e 1.29MB / 1.29MB 2.3s
=> => extracting sha256:a0d8783128af79059fe015e6313ce9e8fd7856b194e57f2d 0.7s
=> => extracting sha256:b79f58b3eb945fe861ace4fc38504483c5b100c794aa095b 2.1s
=> => extracting sha256:48579d87405c21ba2599e244fcf3a02e900dbf9b9103e4af 0.1s
=> [internal] load build context                                   0.2s
=> => transferring context: 470B                                    0.0s
=> [2/5] RUN useradd --create-home appuser                        1.6s
=> [3/5] WORKDIR /app                                             0.1s
=> [4/5] COPY app.py .                                           0.1s

```

5. Run and Test the Image

The image was run as a container using:

```
sudo docker run -d --name day11-web-container -p 8000:8000 day11-web:1.0
```

The running container was verified using:

```
sudo docker ps
```

The application was exposed through:

```
http://localhost:8000
```

The web application successfully displayed:

Day 11 Dockerfile Web App

This confirms that the Docker image was successfully built, started, and tested.

```
shayan@shayan-VirtualBox:~/day11-dockerfile$ sudo docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS
PORTS         NAMES
cb1511add478   day11-web:1.0  "python app.py"         9 seconds ago Up 8 sec
0.0.0.0:8000->8000/tcp, [::]:8000->8000/tcp  day11-web-container
cd675b8f7c57   nginx         "/docker-entrypoint...." 3 hours ago   Up 3 hou
80/tcp        volume-test
shayan@shayan-VirtualBox:~/day11-dockerfile$
```

6. Dockerfile Used

For reference, the final Dockerfile was:

FROM python:3.12-slim

RUN useradd --create-home appuser

WORKDIR /app

COPY app.py .

RUN chown -R appuser:appuser /app

USER appuser

EXPOSE 8000

CMD ["python", "app.py"]

7. Hands-on Task Results

Supervisor Requirement	Result
Containerize a small web app	Completed
Use a non-root user	Completed

Supervisor Requirement	Result
Reduce build context	Completed using .dockerignore
Test the image	Completed